

A Fourier–Compactness Proof of Erdős Problem 42

Corrected version

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Abstract

For every fixed integer $M \geq 1$ we prove that, for all sufficiently large N , every non-empty Sidon set $A \subseteq \{1, \dots, N\}$ admits a Sidon set $B \subseteq \{1, \dots, N\}$ of size M such that

$$(A - A) \cap (B - B) = \{0\}.$$

The proof reduces the assertion to a finite Fourier compactness lemma. The compactness argument passes from cyclic groups of prime order to a compact connected abelian limit group. A positive-definiteness endpoint then forces a positive density of independent tuples.

1 Introduction and notation

A set S of integers is called a Sidon set if

$$s_1 + s_2 = s_3 + s_4, \quad s_i \in S,$$

implies $\{s_1, s_2\} = \{s_3, s_4\}$. Equivalently, every non-zero ordered difference $s - s'$ with $s, s' \in S$ is represented by at most one ordered pair (s, s') .

Theorem 1.1. *For every integer $M \geq 1$ there is $N_0(M)$ such that the following holds for every $N \geq N_0(M)$. If $A \subseteq \{1, \dots, N\}$ is a non-empty Sidon set, then there is a Sidon set $B \subseteq \{1, \dots, N\}$ with $|B| = M$ such that*

$$(A - A) \cap (B - B) = \{0\}.$$

Equivalently, if the non-emptiness assumption on A is removed, the same proof gives

$$(A - A) \cap (B - B) \subseteq \{0\};$$

equality is then possible precisely when A is non-empty.

For a prime p , write $G_p = \mathbb{Z}/p\mathbb{Z}$. For $h : G_p \rightarrow \mathbb{C}$ define

$$\widehat{h}(\xi) = \mathbb{E}_{x \in G_p} h(x) e^{-2\pi i \xi x / p}, \quad \xi \in G_p.$$

Then

$$h(x) = \sum_{\xi \in G_p} \widehat{h}(\xi) e^{2\pi i \xi x / p}, \quad \sum_{\xi \in G_p} |\widehat{h}(\xi)|^2 = \mathbb{E}_{x \in G_p} |h(x)|^2.$$

For a compact abelian group Z with Haar probability measure μ and discrete character group $\Gamma = \widehat{Z}$, groups are written additively and characters multiplicatively. Thus $(\gamma + \gamma')(z) = \gamma(z)\gamma'(z)$ and $(-\gamma)(z) = \overline{\gamma(z)}$. We use the convention

$$\widehat{h}(\gamma) = \int_Z h(z) \overline{\gamma(z)} d\mu(z), \quad \gamma \in \Gamma.$$

2 The finite Fourier compactness lemma

Lemma 2.1 (Fourier independent tuple lemma). *Fix $m \geq 1$, $\alpha > 0$, and $\rho > 0$. There are constants*

$$\varepsilon = \varepsilon(m, \alpha, \rho) > 0, \quad c = c(m, \alpha, \rho) > 0, \quad p_0 = p_0(m, \alpha, \rho),$$

such that the following holds for every prime $p \geq p_0$. Let $F, U \subseteq G_p$ satisfy

$$F = -F, \quad 0 \in F, \quad |F| \leq (1 - \rho)p, \quad |U| \geq \alpha p,$$

and

$$\widehat{\mathbf{1}_F}(\xi) \geq -\varepsilon \quad (\xi \in G_p).$$

Then

$$\#\{(x_1, \dots, x_m) \in U^m : x_i - x_j \notin F \text{ for all } i \neq j\} \geq cp^m.$$

Proof. Suppose the lemma is false. Then for each n there are a prime $p_n \geq n$, sets $F_n, U_n \subseteq G_n := \mathbb{Z}/p_n\mathbb{Z}$, and $\varepsilon_n = 1/n$, such that

$$F_n = -F_n, \quad 0 \in F_n, \quad |F_n| \leq (1 - \rho)p_n, \quad |U_n| \geq \alpha p_n, \\ \widehat{\mathbf{1}_{F_n}}(\xi) \geq -\varepsilon_n \quad (\xi \in G_n),$$

but the normalized count of the desired tuples is less than $1/n$. Put

$$f_n = \mathbf{1}_{F_n}, \quad u_n = \mathbf{1}_{U_n}.$$

Since $F_n = -F_n$, the condition $x_i - x_j \notin F_n$ for all $i \neq j$ is equivalent to the same condition for all $i < j$. Hence

$$\Lambda_n := \mathbb{E}_{x_1, \dots, x_m \in G_n} \prod_{i=1}^m u_n(x_i) \prod_{1 \leq i < j \leq m} (1 - f_n(x_i - x_j)) \longrightarrow 0. \quad (1)$$

Passing to a subsequence, assume also that

$$\mathbb{E}_{G_n} u_n = |U_n|/p_n \longrightarrow \beta \geq \alpha. \quad (2)$$

We shall construct a compact connected abelian group Z and measurable functions $f, u : Z \rightarrow [0, 1]$ such that

$$\Lambda_n \longrightarrow \Lambda := \int_{Z^m} \prod_{i=1}^m u(z_i) \prod_{1 \leq i < j \leq m} (1 - f(z_i - z_j)) d\mu(z_1) \cdots d\mu(z_m).$$

The endpoint argument will show $\Lambda > 0$, contradicting (1).

Step 1: construction of the limiting compact group

For $r \geq 1$ set $\eta_r = 2^{-r}$ and define the large spectrum

$$S_{n,r} = \{\xi \in G_n : |\widehat{f_n}(\xi)| > \eta_r\}.$$

By Parseval,

$$|S_{n,r}| \eta_r^2 \leq \sum_{\xi \in G_n} |\widehat{f_n}(\xi)|^2 = \mathbb{E}_{G_n} |f_n|^2 \leq 1,$$

so $|S_{n,r}| \leq \eta_r^{-2}$. Since $S_{n,r} \subseteq S_{n,r+1}$, a diagonal subsequence allows us to label the spectra compatibly by finite sets

$$A_1 \subseteq A_2 \subseteq \cdots$$

such that, for each fixed r and all sufficiently large n ,

$$S_{n,r} = \{a_n : a \in A_r\},$$

with $a \mapsto a_n$ injective on A_r .

Let $A_* := \bigcup_r A_r$, and let $\mathbb{Z}^{(A_*)}$ be the free abelian group on A_* . For $q \in \mathbb{Z}^{(A_*)}$, let $q_n \in G_n$ denote the corresponding integral linear combination of the labelled frequencies. Since $\mathbb{Z}^{(A_*)}$ is countable, after passing to a further diagonal subsequence we may assume that for every $q \in \mathbb{Z}^{(A_*)}$ either $q_n = 0$ for all sufficiently large n , or $q_n \neq 0$ for all sufficiently large n . Set

$$R = \{q \in \mathbb{Z}^{(A_*)} : q_n = 0 \text{ for all sufficiently large } n\}, \quad \Gamma = \mathbb{Z}^{(A_*)}/R.$$

For $\gamma \in \Gamma$, choose a representative $q \in \mathbb{Z}^{(A_*)}$ and write $\gamma_n = q_n$ for all sufficiently large n . This is well-defined up to changing finitely many n .

For every finite set $C \subseteq \Gamma$, the map $\gamma \mapsto \gamma_n$ is injective on C for all sufficiently large n : if $\gamma \neq \gamma'$, then $\gamma - \gamma' \notin R$, so $(\gamma - \gamma')_n \neq 0$ eventually.

The group Γ is torsion-free. Indeed, if $k\gamma = 0$ in Γ with $k \neq 0$, then $k\gamma_n = 0$ eventually. Since $p_n > |k|$ eventually, multiplication by k is invertible on G_n , whence $\gamma_n = 0$ eventually and $\gamma = 0$.

Let

$$Z = \widehat{\Gamma}.$$

Since Γ is torsion-free, Z is compact and connected. Passing to one more diagonal subsequence, assume that for every $\gamma \in \Gamma$ the limits

$$a_\gamma := \lim_{n \rightarrow \infty} \widehat{f}_n(\gamma_n), \quad b_\gamma := \lim_{n \rightarrow \infty} \widehat{u}_n(\gamma_n)$$

exist. Since $F_n = -F_n$, all $\widehat{f}_n(\xi)$ are real and $\widehat{f}_n(-\xi) = \widehat{f}_n(\xi)$. The lower Fourier bound therefore gives

$$a_\gamma \geq 0 \quad (\gamma \in \Gamma), \tag{3}$$

and also $a_{-\gamma} = a_\gamma$.

Step 2: construction of the limit functions

This is the sign-sensitive part of the proof. Let

$$P(z) = \sum_{\gamma \in C} c_\gamma \overline{\gamma(z)}$$

be a trigonometric polynomial on Z , where $C \subseteq \Gamma$ is finite. For all sufficiently large n , define

$$P_n(x) = \sum_{\gamma \in C} c_\gamma e^{-2\pi i \gamma_n x / p_n} \quad (x \in G_n).$$

For fixed C , the assignment $\gamma \mapsto \gamma_n$ respects all additive relations among elements of C for all sufficiently large n . Indeed, the subgroup of relations among C is finitely generated, and each relation lies in R . Hence, for each $x \in G_n$, the map

$$\gamma \mapsto e^{2\pi i \gamma_n x / p_n}$$

defines a character on the subgroup of Γ generated by C . Since characters on subgroups of discrete abelian groups extend to all of Γ , it is the restriction of some point of $Z = \hat{\Gamma}$. Consequently, if $P \geq 0$ on Z , then $P_n \geq 0$ on G_n for all sufficiently large n .

Moreover,

$$\mathbb{E}_{x \in G_n} P_n(x) \longrightarrow \int_Z P d\mu,$$

because every non-trivial $\gamma \in \Gamma$ satisfies $\gamma_n \neq 0$ eventually.

Define linear functionals on trigonometric polynomials by

$$L_f(P) = \lim_{n \rightarrow \infty} \mathbb{E}_{x \in G_n} f_n(x) P_n(x), \quad L_u(P) = \lim_{n \rightarrow \infty} \mathbb{E}_{x \in G_n} u_n(x) P_n(x).$$

The limits exist by the construction of a_γ and b_γ . If $P \geq 0$, then

$$0 \leq L_f(P) \leq \int_Z P d\mu, \quad 0 \leq L_u(P) \leq \int_Z P d\mu.$$

Thus L_f and L_u are positive bounded linear functionals on the dense $*$ -algebra of trigonometric polynomials, and they extend uniquely to positive linear functionals on $C(Z)$.

The domination by Haar measure also extends to every non-negative continuous function. One way to see this is to convolve with positive trigonometric approximate identities. Such identities exist on Z : for every finite $B \subseteq \Gamma$ and $\tau > 0$, choose a finite non-empty $Q \subseteq \langle B \rangle$ with

$$\frac{|Q \cap (Q - \gamma)|}{|Q|} \geq 1 - \tau \quad (\gamma \in B),$$

and use the Fejer kernel

$$K_Q(z) = \frac{1}{|Q|} \left| \sum_{\eta \in Q} \eta(z) \right|^2.$$

These kernels are non-negative trigonometric polynomials of integral 1, and along an exhaustion of Γ they converge weakly to the point mass at 0. Hence convolution by them approximates continuous functions uniformly.

By the Riesz representation theorem and the Radon–Nikodym theorem, there are measurable functions $f, u : Z \rightarrow [0, 1]$ such that

$$L_f(P) = \int_Z f P d\mu, \quad L_u(P) = \int_Z u P d\mu$$

for every trigonometric polynomial P . Taking $P = \bar{\gamma}$ gives the corrected Fourier identifications

$$\hat{f}(\gamma) = a_\gamma, \quad \hat{u}(\gamma) = b_\gamma \quad (\gamma \in \Gamma). \quad (4)$$

In particular,

$$\int_Z f d\mu = \lim_{n \rightarrow \infty} \mathbb{E}_{G_n} f_n \leq 1 - \rho, \quad \int_Z u d\mu = \lim_{n \rightarrow \infty} \mathbb{E}_{G_n} u_n = \beta \geq \alpha, \quad (5)$$

and by (3),

$$\hat{f}(\gamma) \geq 0 \quad (\gamma \in \Gamma). \quad (6)$$

Step 3: convergence of the finite pattern counts

For $E \subseteq \binom{[m]}{2}$, define

$$T_{E,n}(h, v) = \mathbb{E}_{x_1, \dots, x_m \in G_n} \prod_{i=1}^m v(x_i) \prod_{ij \in E} h(x_i - x_j),$$

where an edge ij is always written with $i < j$. For functions on Z , define

$$T_E(h, v) = \int_{Z^m} \prod_{i=1}^m v(z_i) \prod_{ij \in E} h(z_i - z_j) d\mu(z_1) \cdots d\mu(z_m).$$

We prove that

$$T_{E,n}(f_n, u_n) \longrightarrow T_E(f, u) \quad \left(E \subseteq \binom{[m]}{2} \right). \quad (7)$$

Choose finite sets $B_r \subseteq \Gamma$ such that

$$A_r \subseteq B_r, \quad B_r = -B_r, \quad 0 \in B_r, \quad B_r \subseteq B_{r+1}, \quad \bigcup_r B_r = \Gamma.$$

Since each finite subset of Γ generates a finitely generated torsion-free abelian group, for each r we may choose a finite non-empty set $Q_r \subseteq \Gamma$ such that

$$\frac{|Q_r \cap (Q_r - \gamma)|}{|Q_r|} \geq 1 - \eta_r \quad (\gamma \in B_r). \quad (8)$$

Define

$$K_{n,r}(x) = \frac{1}{|Q_r|} \left| \sum_{\gamma \in Q_r} e^{2\pi i \gamma_n x / p_n} \right|^2 \quad (x \in G_n).$$

For fixed r and all sufficiently large n , the map $\gamma \mapsto \gamma_n$ is injective on $Q_r - Q_r$. Hence

$$K_{n,r} \geq 0, \quad \mathbb{E}_{G_n} K_{n,r} = 1,$$

and all Fourier coefficients of $K_{n,r}$ lie in $[0, 1]$. More precisely, for $\delta \in Q_r - Q_r$,

$$\widehat{K}_{n,r}(\delta_n) = \frac{|Q_r \cap (Q_r - \delta)|}{|Q_r|}. \quad (9)$$

Let

$$g_{n,r} = f_n * K_{n,r}, \quad (a * b)(x) = \mathbb{E}_{y \in G_n} a(x - y)b(y).$$

Then $0 \leq g_{n,r} \leq 1$, and

$$\widehat{g}_{n,r}(\xi) = \widehat{f}_n(\xi) \widehat{K}_{n,r}(\xi).$$

For fixed r and all sufficiently large n ,

$$\sup_{\xi \in G_n} \left| \widehat{f}_n(\xi) - \widehat{g}_{n,r}(\xi) \right| \leq \eta_r. \quad (10)$$

Indeed, if $\left| \widehat{f}_n(\xi) \right| > \eta_r$, then $\xi = \gamma_n$ for some $\gamma \in A_r \subseteq B_r$, and (8)–(9) imply $1 - \widehat{K}_{n,r}(\xi) \leq \eta_r$. If $\left| \widehat{f}_n(\xi) \right| \leq \eta_r$, then (10) follows from $0 \leq \widehat{K}_{n,r}(\xi) \leq 1$.

For $h : G_n \rightarrow \mathbb{C}$ define the Cayley cut norm

$$\|h\|_{\square,n} := \sup_{|a|,|b|\leq 1} |\mathbb{E}_{x,y\in G_n} h(x-y)a(x)b(y)|.$$

Fourier inversion and Cauchy–Schwarz give

$$\|h\|_{\square,n} \leq \sup_{\xi \in G_n} |\widehat{h}(\xi)|. \quad (11)$$

Consequently,

$$\|f_n - g_{n,r}\|_{\square,n} \leq \eta_r. \quad (12)$$

A one-edge telescoping argument gives

$$|T_{E,n}(f_n, u_n) - T_{E,n}(g_{n,r}, u_n)| \leq |E| \eta_r. \quad (13)$$

Indeed, when one edge ij is replaced by $f_n - g_{n,r}$ and all other variables are fixed, the remaining factors involving x_i and x_j are bounded by 1 and factor as a function of x_i times a function of x_j . Thus (12) applies.

For fixed r , set $D_r = Q_r - Q_r$. For all sufficiently large n , the Fourier support of $g_{n,r}$ is contained in $(D_r)_n$, and

$$g_{n,r}(x) = \sum_{\delta \in D_r} \widehat{f}_n(\delta_n) \widehat{K}_{n,r}(\delta_n) e^{2\pi i \delta_n x / p_n}.$$

Expanding the finite Fourier series in $T_{E,n}(g_{n,r}, u_n)$, each term is indexed by a map $\omega : E \rightarrow D_r$. Define

$$\sigma_i(\omega) = \sum_{j:i < j, ij \in E} \omega(ij) - \sum_{j:j < i, ji \in E} \omega(ji).$$

The corresponding contribution is

$$\prod_{ij \in E} \widehat{f}_n(\omega(ij)_n) \widehat{K}_{n,r}(\omega(ij)_n) \prod_{i=1}^m \widehat{u}_n(-\sigma_i(\omega)_n).$$

The finite set consisting of D_r and all sums $\sigma_i(\omega)$ is fixed once r and E are fixed, so the construction of Γ gives eventual injectivity and all required coefficient limits. Therefore

$$\lim_{n \rightarrow \infty} T_{E,n}(g_{n,r}, u_n) = T_E(g_r, u), \quad (14)$$

where

$$K_r(z) = \frac{1}{|Q_r|} \left| \sum_{\gamma \in Q_r} \gamma(z) \right|^2, \quad g_r = f * K_r.$$

Here convolution on Z is with respect to Haar probability measure.

The kernels K_r form a positive approximate identity on Z . They satisfy $K_r \geq 0$ and $\int_Z K_r d\mu = 1$, and for every fixed $\gamma \in \Gamma$,

$$\widehat{K_r}(\gamma) \rightarrow 1,$$

because B_r exhausts Γ and Q_r is Følner for B_r . Thus the probability measures $K_r d\mu$ converge weakly to the point mass at $0 \in Z$. Hence, for every $f \in L^1(Z)$,

$$\|f * K_r - f\|_1 \leq \int_Z \|\tau_y f - f\|_1 K_r(y) d\mu(y) \rightarrow 0,$$

where $\tau_y f(z) = f(z - y)$. This follows from L^1 -continuity of translations and weak concentration of $K_r d\mu$ at 0. In particular,

$$g_r \rightarrow f \quad \text{in } L^1(Z). \quad (15)$$

Since $0 \leq f, g_r, u \leq 1$, (15) implies

$$T_E(g_r, u) \longrightarrow T_E(f, u). \quad (16)$$

Combining (13), (14), and (16), and then letting $r \rightarrow \infty$, proves (7).

By inclusion-exclusion,

$$\Lambda_n = \sum_{E \subseteq \binom{[m]}{2}} (-1)^{|E|} T_{E,n}(f_n, u_n).$$

Therefore (7) gives

$$\Lambda_n \longrightarrow \Lambda = \int_{Z^m} \prod_{i=1}^m u(z_i) \prod_{1 \leq i < j \leq m} (1 - f(z_i - z_j)) d\mu^m. \quad (17)$$

By (1), this gives $\Lambda = 0$.

Step 4: the positive-definite endpoint

We now show that $\Lambda > 0$, contradicting (17). Recall that

$$0 \leq f \leq 1, \quad \int_Z f d\mu \leq 1 - \rho, \quad \widehat{f}(\gamma) \geq 0 \quad (\gamma \in \Gamma).$$

First we prove that f has an absolutely convergent Fourier series. Let $C \subseteq \Gamma$ be finite and let $\tau > 0$. Choose a finite non-empty set $Q \subseteq \Gamma$ such that

$$\frac{|Q \cap (Q - \gamma)|}{|Q|} \geq 1 - \tau \quad (\gamma \in C \cup (-C)). \quad (18)$$

Set

$$K_Q(z) = \frac{1}{|Q|} \left| \sum_{\eta \in Q} \eta(z) \right|^2.$$

Then $K_Q \geq 0$, $\int_Z K_Q d\mu = 1$, and

$$\widehat{K}_Q(\gamma) = \frac{|Q \cap (Q - \gamma)|}{|Q|}.$$

Since $0 \leq f \leq 1$,

$$1 \geq \int_Z f K_Q d\mu = \sum_{\gamma \in \Gamma} \widehat{f}(\gamma) \widehat{K}_Q(-\gamma) \geq (1 - \tau) \sum_{\gamma \in C} \widehat{f}(\gamma),$$

where the sum is finite because K_Q is a trigonometric polynomial and (18) controls $-\gamma$ for $\gamma \in C$. Letting $\tau \downarrow 0$ gives

$$\sum_{\gamma \in C} \widehat{f}(\gamma) \leq 1.$$

Since C was arbitrary and all coefficients are non-negative,

$$\sum_{\gamma \in \Gamma} \widehat{f}(\gamma) \leq 1. \quad (19)$$

Thus

$$F(z) = \sum_{\gamma \in \Gamma} \widehat{f}(\gamma) \gamma(z)$$

converges absolutely and uniformly. It has the same Fourier coefficients as f , so $F = f$ almost everywhere. Replacing f by this continuous representative, the bounds $0 \leq f \leq 1$ still hold everywhere, since Haar measure has full support.

Let

$$H = \{z \in Z : f(z) = 1\}.$$

We claim that

$$\mu(H) = 0. \tag{20}$$

If $H = \emptyset$, this is immediate. Otherwise choose $h_0 \in H$. By (19), the vector

$$v = (\sqrt{\widehat{f}(\gamma)})_{\gamma \in \Gamma}$$

belongs to $\ell^2(\Gamma)$. Let U_z be the unitary representation of Z on $\ell^2(\Gamma)$ given by

$$U_z e_\gamma = \gamma(z) e_\gamma.$$

With the inner product linear in the first variable,

$$f(z) = \langle U_z v, v \rangle.$$

Since $h_0 \in H$,

$$1 = f(h_0) = \langle U_{h_0} v, v \rangle.$$

Cauchy–Schwarz gives

$$1 = |\langle U_{h_0} v, v \rangle| \leq \|U_{h_0} v\| \|v\| = \|v\|^2 = f(0) \leq 1.$$

Thus $f(0) = 1$, $\|v\| = 1$, and equality holds in Cauchy–Schwarz. Since $\langle U_{h_0} v, v \rangle = 1$, equality forces $U_{h_0} v = v$. More generally, for any $h \in Z$,

$$f(h) = 1 \iff U_h v = v.$$

Therefore H is the stabilizer of v , and hence is a closed subgroup of Z .

A closed subgroup of a compact group with positive Haar measure is open. Since Z is connected, it has no proper open subgroup. Therefore, if $\mu(H) > 0$, then $H = Z$, which would imply $f \equiv 1$ and hence $\int_Z f d\mu = 1$, contradicting (5). This proves (20).

Now set

$$S = \{z \in Z : u(z) > 0\}.$$

By (5), $\mu(S) > 0$. For each pair $i < j$, Haar invariance and Fubini's theorem give

$$\mu^m\{(z_1, \dots, z_m) \in S^m : z_i - z_j \in H\} = 0.$$

Hence, for almost every $(z_1, \dots, z_m) \in S^m$,

$$z_i - z_j \notin H \quad (1 \leq i < j \leq m).$$

For such tuples,

$$u(z_i) > 0 \quad (1 \leq i \leq m), \quad 1 - f(z_i - z_j) > 0 \quad (1 \leq i < j \leq m).$$

Thus the integrand defining Λ is positive almost everywhere on the positive-measure set S^m . Therefore $\Lambda > 0$, contradicting $\Lambda = 0$.

This contradiction proves the lemma. $\square$

3 Application to Sidon sets

Proof of Theorem 1.1. Fix $M \geq 1$. Apply Lemma 2.1 with

$$m = M, \quad \alpha = \frac{1}{8}, \quad \rho = \frac{1}{2}.$$

Let the resulting constants be ε_0, c_0, p_0 .

Let N be sufficiently large in terms of M , and let $A \subseteq \{1, \dots, N\}$ be a non-empty Sidon set. Put $k = |A|$. By Bertrand's postulate, choose a prime

$$4N < p < 8N.$$

We embed $\{1, \dots, N\}$ into G_p . Since $p > 2N$, all differences between elements of $\{1, \dots, N\}$ are represented modulo p without wraparound.

Let

$$F = A - A \subseteq G_p.$$

Then $F = -F$ and $0 \in F$. Since A is Sidon, every non-zero ordered difference $a - a'$ with $a, a' \in A$ is represented at most once. Hence

$$|F| = 1 + k(k-1).$$

Also the positive differences of A are distinct and belong to $\{1, \dots, N-1\}$, so

$$\binom{k}{2} \leq N-1.$$

Therefore

$$|F| = 1 + k(k-1) \leq 2N-1 < \frac{p}{2}. \quad (21)$$

Next we verify the Fourier lower bound. For $t \in G_p$, let

$$r(t) = \#\{(a, a') \in A^2 : a - a' = t\}.$$

Then $r(0) = k$, and by the Sidon property $r(t) = \mathbf{1}_F(t)$ for $t \neq 0$. Hence

$$\mathbf{1}_F = r - (k-1)\mathbf{1}_{\{0\}}.$$

For every $\xi \in G_p$,

$$\widehat{r}(\xi) = \frac{1}{p} \left| \sum_{a \in A} e^{-2\pi i \xi a / p} \right|^2 \geq 0,$$

and therefore

$$\widehat{\mathbf{1}}_F(\xi) = \widehat{r}(\xi) - \frac{k-1}{p} \geq -\frac{k-1}{p}. \quad (22)$$

Since $\binom{k}{2} \leq N-1$, we have $k = O(\sqrt{N})$, while $p > 4N$. Thus, by taking N sufficiently large in terms of M , (22) gives

$$\widehat{\mathbf{1}}_F(\xi) \geq -\varepsilon_0 \quad (\xi \in G_p),$$

and also $p \geq p_0$.

Let

$$U = \{1, \dots, N\} \subseteq G_p.$$

Since $p < 8N$, we have $|U| = N > p/8$. By Lemma 2.1 and (21), there are at least $c_0 p^M$ ordered tuples

$$(x_1, \dots, x_M) \in U^M$$

such that

$$x_i - x_j \notin F \quad (i \neq j). \quad (23)$$

Because $0 \in F$, every tuple satisfying (23) has distinct entries.

We now discard tuples whose non-zero ordered differences are not all distinct. For two distinct ordered pairs $(i, j) \neq (r, s)$ with $i \neq j$ and $r \neq s$, the relation

$$x_i - x_j = x_r - x_s$$

is a non-trivial linear equation in $x_1, \dots, x_M$. It has $O_M(N^{M-1})$ solutions in U^M . Since there are $O_M(1)$ choices of the two ordered pairs, the total number of tuples in U^M with a repeated non-zero ordered difference is

$$O_M(N^{M-1}). \quad (24)$$

Taking N still larger if necessary, (24) is smaller than $c_0 p^M$. Therefore at least one tuple satisfying (23) has all non-zero ordered differences distinct.

Choose such a tuple and set

$$B = \{x_1, \dots, x_M\} \subseteq \{1, \dots, N\}.$$

The entries are distinct, so $|B| = M$. Since all non-zero ordered differences of the chosen tuple are distinct, B is Sidon. Finally, if a non-zero integer difference of B belonged to $A - A$, then its residue modulo p would lie in F , contradicting (23). Thus

$$(A - A) \cap (B - B) = \{0\},$$

because A and B are non-empty. □

Remark 3.1. *The non-emptiness assumption on A is only needed for the final equality with $\{0\}$. If $A = \emptyset$, then $(A - A) \cap (B - B) = \emptyset$ for every B , so the equality with $\{0\}$ is impossible. The proof above still gives the natural avoidance statement $(A - A) \cap (B - B) \subseteq \{0\}$.*

References

- [1] T. F. Bloom, *Erdős Problem #42*, <https://www.erdosproblems.com/42>, accessed April 27, 2026.